

BUILD WITH AI: CODE FOR COMMUNITIES · TRACK 5

ShadowCast

Names the hospitals, shelters and substations a cyclone will knock out, then checks itself against satellite data.

PILOT: CYCLONE FANI × ODISHA COAST
TEAM ARGMAX · SOLO BUILD

shadowcast-two.vercel.app · github.com/tsathya98/shadowcast



Puri had no mains power 11 days after Fani

1.2 M

people evacuated to shelters
before landfall on 3 May 2019

11 days

later, Puri district still had no
mains power

3,325

hospitals, shelters, substations
and other critical sites on this
coast

We build for the district emergency officer at T-48 h, who decides which sites get generators, line crews and warnings first.

Warnings describe the storm; officers need a ranked list of sites

What a warning gives	What the officer needs
Track, cone and landfall point	Which named sites are at risk
Wind category by district	When gales reach each site
One best-guess track	How likely each site is to lose power
No look back afterwards	Proof the model was right last time

In Fani, all 448 assets in Puri were inside the 64-knot wind radius, so a band map gives them all the same colour. Modelled wind at each site is what separates them.

THE SOLUTION

Know which sites a cyclone will knock out, and when, up to 68 hours ahead

RANK #1 OF 3,325 · FANI 2019 REPLAY

Mot cyclone shelter, Brahmagiri

99% chance of losing power

- Peak wind 126 kt · storm passes 9 km away
- Gales from 2 May 16:30 IST, 16 h before landfall
- 8,219 people within 2 km

Alert drafted in English, Hindi, Odia · awaiting approval



01 PREDICT

ECMWF forecasts as issued: wind, surge, rain, roads

02 PRIORITISE

Model trained on outages seen by satellite, × criticality

03 PREPARE

Gemini drafts; officer signs

04 PROVE

Satellites score every storm; misses published

THE LIVE PRODUCT

The live console, two hours before Fani's landfall



- 01 Read the brief
IMD's bulletin, live warnings, and each agency's deadline
- 02 See what breaks
Power, surge, rain and roads closing, site by site
- 03 Issue the alert
Gemini drafts; approving publishes it on the CAP feed
- 04 Check afterwards
Satellites and Gemini score every storm

Gemini does the reading and drafting; an officer approves every alert

IMD BULLETIN (PDF)

IMD's official bulletin, straight from the PDF: Fani's 1.5 m surge over Ganjam, Khurda, Puri, Jagatsinghpur

SATELLITE IMAGES

Before/after VIIRS night lights: "Khordha, Puri and Cuttack totally dark", which agrees with the forecast

LIVE VOICE CALLS

Call the analyst over Gemini Live in any Indian language; it answers aloud and can be interrupted

TOOL CALLS

Why a site ranks where it does, from live calls to the geo API and IMD's bulletin

CAP 1.2 ALERTS

The public alert in English, Hindi and the local language; once approved, it is published on the CAP feed

TEXT TO SPEECH

Gemini TTS reads approved alerts and chat replies aloud, in six languages

Gemini never sets a probability (the trained outage model does), never sees the outcome during a forecast replay, and issues nothing without the officer's signed approval. Every decision goes to an append-only Firestore audit log, and IMD remains the authority.

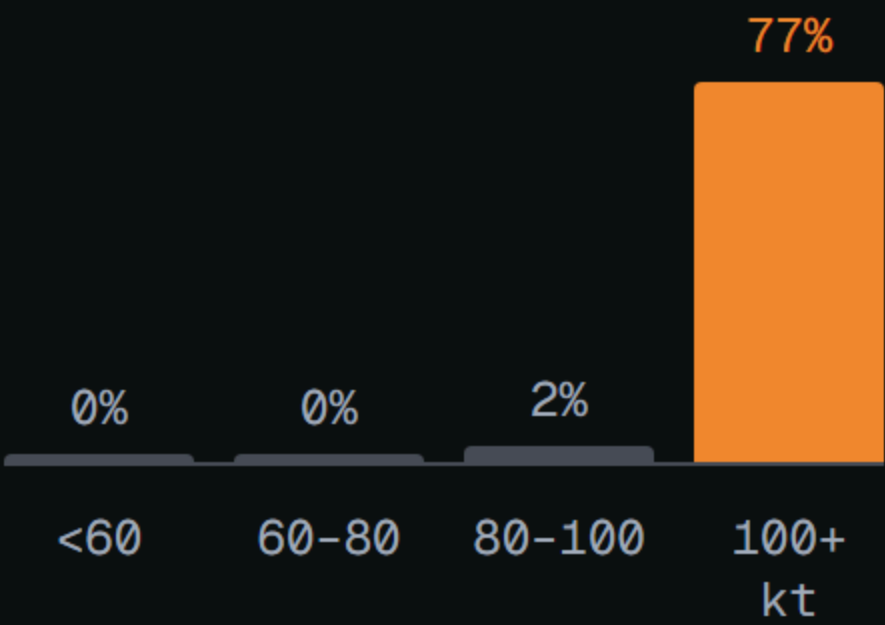
Public satellite, weather and map data

Layer	Source	Used for
Tracks, forecasts	IBTrACS (NOAA); ECMWF ensemble as issued	Wind, rain; 36–52 members
Official warnings	IMD bulletins (PDF); NDMA SACHET; GDACS	Read by Gemini; live feed
Sites, roads	OpenStreetMap; OSDMA cyclone shelters	3,325 sites; arterial roads
People, districts	WorldPop; geoBoundaries (Earth Engine)	Exposure; parametric cover
Ground, sea floor	DeltaDTM, Copernicus DEM, ETOPO1	Surge flooding
Truth	NASA VIIRS night lights; NASA GPM rain	Which sites went dark; rain

Everything is pulled through Earth Engine or open APIs and credited in the repository. None of it is synthetic.

Every model scored against what really happened

Fani: median night-light loss at substations, by modelled peak wind



Storm	Outage AUC	Surge vs IMD	Rain ρ
Fani 2019	0.97 · 0.97*	2.3 vs 1.5 m	0.72
Hudhud 2014	0.79	1.7 vs 1.4 m	0.85
Amphan 2020	0.44 fails	3.8 vs 4.6 m	0.14
Dana 2024	5 outages	2.8 vs 1–2 m	0.96

Outage truth: NASA VIIRS night lights. Surge: IMD's reported height (Hudhud from a tide gauge). Rain: rank correlation with NASA GPM.
*Spatial holdout.

SATELLITE EVIDENCE · READ BY GEMINI

BEFORE · VIIRS

AFTER · VIIRS

AGREES WITH SHADOWCAST

The post-landfall image reveals widespread power loss across coastal and central Odisha, most notably extinguishing the major urban light footprints in Khordha, Cuttack, and Puri. In contrast, inland industrial centers to the northwest and coastal clusters to the northeast and southwest remained largely intact.

Khordha · total blackout · The bright metropolitan light center across Khordha completely collapsed into darkness.

Puri · total blackout · Coastal lighting in Puri vanished entirely after landfall.

Cuttack · total blackout · Lights directly north of Khordha in Cuttack were almost completely extinguished.

Amphan fails on outages and rain: wind barely varies over land in the model and the rural grid failed widely. A physics fix made Fani worse (0.87), so we reverted it and publish the miss.

SHADOWCAST · TEAM ARGMAX · FULL RECORD IN THE REPO

08

IMPACT

Hours of warning, per site, road and district

45/57

Fani substations that went dark had been flagged at 50% or more

68 h

before landfall: per-site gale odds from the as-issued ensemble (Dana)

T-4 h

NH316, Bhubaneswar to Puri, becomes unsafe: the deadline for its shelters

18% · 0%

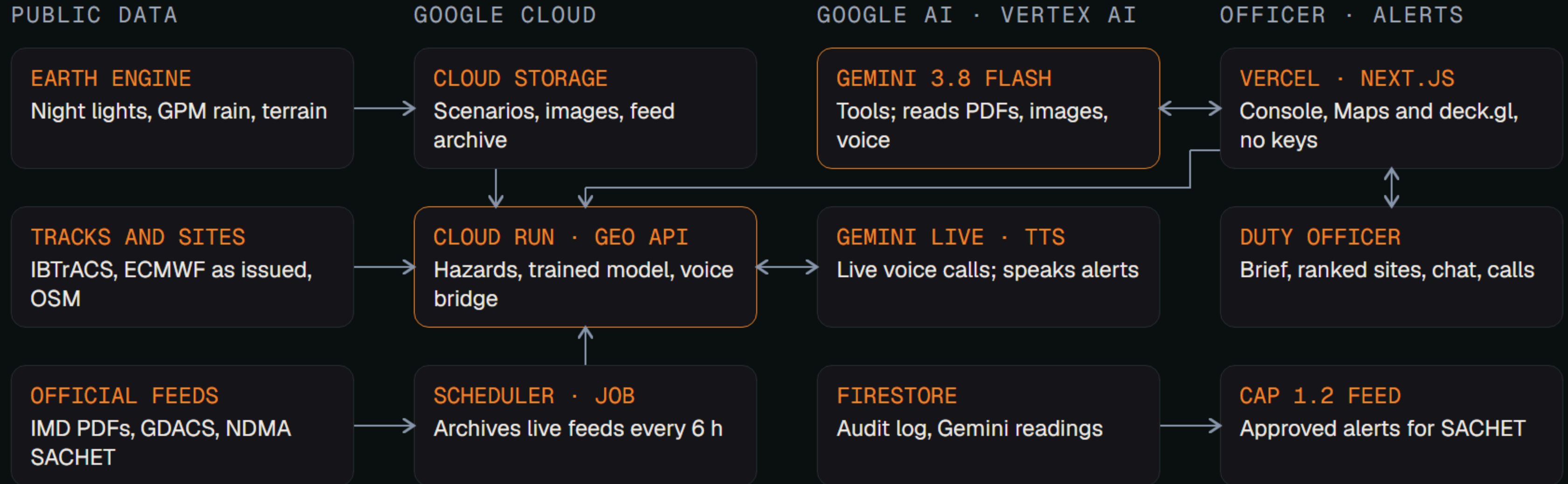
outages where the parametric cover triggered, and where it did not

Proposed pilot success criteria

- At least 70% of sites that lose power flagged beforehand, verified by satellite within a week
- Priority sites and their access roads get 12 h or more of named warning before gales arrive
- Every advisory approved or rejected by an officer, with a complete audit trail

TECHNICAL ARCHITECTURE

Live today on Google Cloud and Vercel



The console reads the geo API and calls Gemini with no stored keys (Workload Identity Federation); voice calls run through the geo service to Gemini Live. 100% line coverage, CI on every push. One Odisha season for 30 officers costs **about ₹5,300 a month**.

Three coasts already in the data

Odisha coast

Fani 2019 · Dana 2024
3,325 sites · alerts in Odia

North Andhra coast

Hudhud 2014
1,729 sites · alerts in Telugu

West Bengal coast

Amphan 2020
2,227 sites · alerts in Bengali

Adding a coast

Every input is global open data. A new coast is a region entry, a language and one build; the outage model is then checked against that grid's own night lights.

Plugging into national alerting

Alerts are CAP 1.2, the format NDMA's SACHET already carries, published on a standard CAP feed. The same build works for any cyclone basin, from India to the cyclone coasts of other BRICS nations.

THE ASK

One cyclone season alongside OSDMA

OCT-DEC 2026

Shadow-run

Run beside IMD bulletins on live ECMWF forecasts for Odisha; verify each storm by satellite within a week.

NEXT

Recalibrate

Fit the outage model per utility grid and add inland wind decay: the Amphan lesson.

THEN

Scale

Andhra and West Bengal disaster authorities; approved alerts out through SACHET.

What we need: read access to one utility's outage logs, and regular feedback from one district officer.